ICS4U
Unit: Algorithms

Algorithm to Code

Bubble Sort Algorithm

Lay a series of unsorted number cards on the table.
Follow the algorithm to practice how a computer processes instructions.

```
do
     swapped = false
     for i = 1 to index_of_last_unsorted_item - 1
          if left_element > right_element
               swap(left_element, right_element)
               swapped = true
while swapped
```

Complete a worked example (later)

Practice by writing out the swaps/iterations to sort a series of numbers using this sort.

Translating Algorithm to Code

```
def bubble_sort(a_list):



    ##  swapped = false

     ##  for i = 1 to index_of_last_unsorted_item - 1



     ##    if left_element > right_element

     ##          swap(left_element, right_element)



     ##          swapped = true









a = [5, 4, 6, 2, 4, 6, 7, 8, 1]
bubble_sort(a)
print(a)
```

```
def other_bubble_sort(a_list):









a = [5, 4, 6, 2, 4, 6, 7, 8, 1]
other_bubble_sort(a)
print(a)
```